SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards *(BL2,BL6)*

Assessment Tool Weightage: 30%

QUESTIONS: 5 **Total Marks:25**

Annexure A **Comparative Analysis**

Code of Conduct:

I __________GODDAVULA NITISH KANTH (192512232) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:NITISH KANTH.G

Annexure A

Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it lowcost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.
2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.
3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.
4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.
5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

1. Comparative Analysis: Star Topology vs Bus Topology

Answer

Star topology and Bus topology are two commonly used network topologies, but they differ significantly in terms of cost, reliability, performance, and maintenance. Star topology connects every computer or device to a central switch or hub using individual cables. This requires more cabling and networking hardware, making the installation cost higher than bus topology. However, the centralized design provides better network management and easier troubleshooting.

In contrast, Bus topology connects all devices through a single backbone cable. Since only one main cable is used, the installation cost is much lower, making it suitable for small networks or classroom laboratories with limited budgets. However, all devices share the same communication medium, so network traffic increases as more devices are connected.

The biggest advantage of star topology is fault isolation. If one computer or cable fails, only that particular device loses connectivity, while the rest of the network continues to work normally. In bus topology, if the backbone cable is damaged or disconnected, the entire network stops functioning. Star topology also provides better performance because switches reduce data collisions and improve communication speed, whereas bus topology suffers from collisions and slower performance during heavy traffic.

Comparison Table

Feature	Star Topology	Bus Topology
Cost	Higher due to switch and more cables.	Lower because only one backbone cable is required.
Reliability	High; failure affects only one node.	Low; backbone cable failure stops the entire network.
Performance	High-speed communication with fewer collisions.	Performance decreases as devices increase.
Maintenance	Easy to identify and repair faults.	Difficult to locate backbone cable faults.

Conclusion

Star topology is ideal for offices and organizations where reliability and performance are important, while bus topology is suitable for small, low-cost networks with limited devices.

2. Comparative Analysis: Mesh Topology vs Star Topology

Answer

Mesh topology is considered the most reliable network topology because every network device is connected to multiple other devices. This creates multiple communication paths, allowing data to travel through alternate routes if one link fails. As a result, mesh topology provides very high fault tolerance, making it suitable for banks, hospitals, military communication systems, and data centers.

Star topology also provides reliable communication, but all devices depend on the central switch or hub. If a workstation cable fails, only that device is affected. However, if the central switch fails, the entire network becomes unavailable until the switch is repaired or replaced.

Mesh topology offers continuous connectivity because routing protocols such as OSPF and BGP automatically redirect traffic through available links during failures. This ensures uninterrupted communication even during hardware or cable failures. However, mesh topology requires a large number of cables and ports, increasing installation cost and network complexity.

Comparison Table

Feature	**Mesh Topology**	**Star Topology**
Fault Tolerance	Very high with multiple backup paths.	Moderate; depends on central switch.
Reliability	Excellent for critical applications.	Good for offices and schools.
Cost	Very expensive.	Moderate installation cost.
Complexity	Highly complex network design.	Simple and easy to manage.

Conclusion

Mesh topology provides maximum network availability and redundancy, while star topology offers a balance between reliability, simplicity, and cost.

3. Comparative Analysis: Bus Topology vs Mesh Topology

Answer

Bus topology and mesh topology represent two opposite approaches to network design. Bus topology is designed for simplicity and low cost, while mesh topology is designed for reliability and uninterrupted communication.

In bus topology, all devices communicate using one shared backbone cable. This makes installation quick and inexpensive because only one cable runs through the network. It is suitable for small classroom labs or temporary office setups where network traffic is low.

Mesh topology requires every device to be connected to every other device (full mesh) or to multiple important devices (partial mesh). This results in many cables, ports, and network interfaces, making installation expensive and time-consuming. As the number of devices increases, the complexity of mesh topology grows rapidly.

Bus topology has poor fault tolerance because damage to the backbone cable disconnects all devices. Mesh topology continues operating even if several links fail because data automatically uses alternate paths

Comparison Table

Feature	**Bus Topology**	**Mesh Topology**
Installation	Simple and easy.	Complex and time-consuming.
Cabling	Very few cables required.	Large number of cables required.
Fault Tolerance	Low.	Very high.
Cost	Low.	Very high.
Best Application	Small classroom or temporary network.	Banks, data centers, enterprise backbone.

Conclusion

Bus topology is economical but less reliable, whereas mesh topology provides uninterrupted communication for mission-critical networks.

4. Comparative Analysis: Hybrid Topology vs Star Topology

Answer

Hybrid topology combines two or more network topologies such as Star, Mesh, Ring, and Tree to create a flexible and scalable network infrastructure. It is commonly used in universities, hospitals, airports, and large organizations with multiple departments or buildings.

Star topology is simple because all devices connect to one central switch. It works efficiently for small and medium-sized organizations, but expansion depends on the number of available switch ports. Once the switch capacity is exhausted, another switch must be installed.

Hybrid topology allows different sections of an organization to use different topologies according to their needs. For example, a university may use mesh topology for the campus backbone and star topology inside each building. This provides high-speed communication, redundancy, and flexibility for future expansion.

Comparison Table

Feature	**Hybrid Topology**	**Star Topology**
Scalability	Very high and flexible.	Moderate; depends on switch ports.
Flexibility	Supports multiple topologies.	Single network structure.
Reliability	High due to redundant backbone.	High for small networks.
Cost	Higher installation cost.	Moderate cost.
Applications	University campuses, enterprises.	Offices, schools, startups.

Conclusion

Hybrid topology is the preferred solution for large organizations because it combines scalability, redundancy, and flexibility, while star topology is ideal for smaller organizations.

5. Comparative Analysis: Maintenance of Network Topologies

Answer

Maintenance is an important factor when selecting a network topology because it affects troubleshooting time, operational cost, and network availability.

Star topology is the easiest to maintain because each device has a dedicated cable connected to the switch. Faults can be quickly identified using switch indicators, cable testers, or network management software. Replacing one cable or device does not interrupt other users.

Mesh topology is the most difficult to maintain because it contains many interconnected links. Although it provides excellent reliability, identifying faulty cables or devices requires careful monitoring and experienced network administrators.

Bus topology is harder to maintain than star topology because all devices depend on one backbone cable. Finding the exact location of a cable break or connector fault can take time, and the entire network remains unavailable until the fault is repaired.

Hybrid topology has moderate maintenance complexity. Since it combines multiple topologies, administrators must monitor different network segments separately. Enterprise monitoring tools are often used to detect and isolate faults quickly.

Maintenance Comparison Table

Topology	Maintenance Level	Reason
Star Topology	Easy	Dedicated cables make fault isolation simple.
Mesh Topology	Difficult	Large number of links increases troubleshooting complexity.
Bus Topology	Moderate to Difficult	Backbone cable faults affect the complete network.
Hybrid Topology	Moderate	Requires monitoring of multiple combined topologies.

Conclusion

Star topology is the easiest and most cost-effective topology to maintain. Meshtopology offers the highest reliability but requires skilled maintenance. Bus topology is difficult because of backbone dependency,whilehybridtopology provides flexibility but needs professional network management tools and administrators.